Linear Regression under Strategic Feature Withholding

Manish Kumar Singh[1], Ankur A. Kulkarni[1], and Kriti Mahajan[2]

Abstract— **We consider the problem of linear regression as a game, in which a sender of data decides whether or not to reveal the feature vector (voluntary reporting) to a receiver based on a private preference for the prediction outcome, and the receiver decides which linear model to use for prediction. If the sender decides to reveal the feature vector, it is communicated through a noisy channel, and there is a fixed probability that the feature vector is lost. If the sender does not reveal the feature vector, one expects the prediction accuracy to deteriorate due to missing information. In our work, we show that this intuition is not always correct. If the receiver accounts for strategic omission, and if there is sufficient alignment between the sender and the receiver, a linear model adapted to such behavior can achieve a strictly lower mean squared error than the best linear model with full feature access. We derive a necessary condition and a sufficient condition under which voluntary reporting yields lower risk. This result has important implications for disclosure policies in self-reporting environments, such as financial markets. When the underlying environment is unknown, we propose a joint selection learning algorithm that simultaneously learns the optimal disclosure regime (mandatory or voluntary) and its corresponding optimal predictor. Notably, the error rate of this algorithm matches that of any standard learning algorithm that knows the optimal disclosure regime a priori.**

I. INTRODUCTION

The standard linear regression model assumes that all relevant features are available at the time of inference. In this work, we study a natural variant in which a sender can selectively report its feature vector to a receiver based on a private objective that is correlated with the ground truth. The receiver must design a linear model to be used for prediction on the feature vector communicated by the sender. The central question we ask is: When is such opportunistic reporting actually beneficial to the receiver in terms of mean squared error (MSE)? To exploit this behavior of the sender, the regression model must explicitly account for deliberate omissions during prediction. If omissions are systematic, for instance, when all senders share the same preference ordering over outcomes, then assigning the worst outcome to the sender in the case of a missing feature vector suffices to guarantee risk at least as low as that of the mandatory disclosure regime. However, when preferences among senders are non-uniform, such simple remedies are no longer sufficient.

We model this interaction as a Stackelberg game between a sender and a receiver. The receiver first commits to a collection of linear regression models, one for each possible subset of features revealed by the sender. A sender is then drawn at random from a population and possesses a feature vector, a preferred value, and a ground-truth value. Upon observing the feature vector, the preferred value, and the receiver's committed strategy, the sender decides which action to take (in our setting, all-or-nothing). If a non-empty message is chosen, it is transmitted through a noisy channel with a fixed, independent probability of omission. This noisy channel introduces uncertainty, as the receiver cannot distinguish between a deliberate omission and a random erasure. The resulting action is observed by the receiver as a message. Depending on the message, the receiver applies the corresponding committed linear regression model to obtain an estimate of the ground-truth value. The receiver incurs an MSE based on the estimate and the ground-truth value, while the sender incurs an MSE based on the estimate and their preferred value.

A concrete example of non-uniform preferences arises in financial markets. A firm (the sender) possesses a ground-truth valuation that the market (the receiver) seeks to estimate from a financial report. Additionally, each firm maintains an internal preferred valuation that may differ across firms. Some firms prefer overvaluation to boost stock prices and lower borrowing costs, while others prefer undervaluation ahead of stock buybacks or to avoid regulatory scrutiny. When voluntary disclosure is permitted, a firm reveals its full feature vector only when doing so moves the market's estimate closer to its preferred valuation. If the preferred valuation is close to the ground-truth valuation, this selective disclosure helps the market by filtering noisy features. If the preferred valuation is far from the ground-truth valuation, the firm withholds precisely the features that are most informative for estimating the ground-truth value. Thus, the market must choose disclosure rules carefully.

A. Main findings

In our work, we focus on the setting in which a feature vector is either revealed in its entirety or kept completely hidden before entering the noisy channel (all-or-nothing disclosure). The receiver commits to two linear regression models (one for "all" and one for "nothing") for prediction. When the environment is known, we derive a necessary condition and a sufficient condition under which the voluntary disclosure regime yields strictly lower mean squared error than the mandatory disclosure regime.

The reason for the lower mean squared error is as follows: We can interpret the feature ground-truth value pair as coming from a mixture of preference values. So if there is sufficient alignment between the sender's preference value

[1]Centre for Systems and Control, Indian Institute of Technology Bombay, India. Email: {204230004, kulkarni.ankur}@iitb.ac.in

[2]Amazon India. Email: kritimhj@amazon.com

and the ground-truth value, then the sender acts as a filter that only lets those components of the mixture pass through that match his preference, which benefits the receiver as those values are highly aligned with the ground-truth value. This explains the counter-intutive result that witholding feature helps the receiver. The sender's revelation is strategic in nature, and under sufficient allignment between the sender preference and the ground-truth value, the sender's action reveals more about the ground truth than the actual feature vector, to the receiver.

When only finite samples are available, we propose a learning algorithm that jointly selects both the linear predictor and the disclosure regime (voluntary or mandatory). We compare the algorithm's risk to that of the optimal predictor under the optimal disclosure regime and establish uniform excess-risk bounds, showing that it is of the same order as the risk achieved by an oracle algorithm that knows the optimal disclosure regime a priori. This happens because the feature-ground-truth value pair can be though of as comming from mixtures of preference values.

B. Related works

Our work lies in the domain of learning with strategic agents. [1] [2] [3] pioneered the approach of formulating the interaction with strategic agents during learning as a sender-receiver game. We also adopt this structure, which has served as the basis for many subsequent works. [4], [5] consider the problem of making their models robust to manipulation by the sender. [6] studied the setting where strategic manipulation results in a change of the causal structure of the sender. [7] analysed the setting where incentives were designed to encourage strategic behaviour that is genuinely beneficial for the sender rather than gaming the system. [8] consider the setup in linear regression, where the sender deliberately provided less precise data to a public model, but expected to benefit from its use. A common theme across all these works is that strategic manipulation by the sender involved altering the feature vector. In contrast, our work considers the setting where strategic manipulation refers to withholding feature vectors in a regression setting. Although [9] also considers strategic manipulation to be feature withholding, their work focuses on a classification problem and assumes a uniform preference setup. Furthermore, works like [10], [11] share with us the non-uniform preference setup, however, they differ in that they were focus on strategic PAC learnability for classification and their strategic manipulation also involved altering the feature vector. Seminal works in economics [12], [13] do consider the problem of voluntary revelation, but these are problems with the sender as leader, which differ from our formulation of the receiver as leader. To the best of our knowledge, the unique combination of strategic manipulation as feature withholding, non-uniform preference, and receiver as a leader has not been studied earlier.

II. Problem Formulation

Let $\mathcal{X} = \mathbb{R}^d$ be the feature space and $\mathcal{Y} = \mathbb{R}$ be the outcome space. A sender observes a tuple (U, X) drawn at random, with zero mean, from the space $\mathcal{Y} \times \mathcal{X}$ and nature selects $Y \in \mathcal{Y}$ with zero mean. Here, X denotes the sender's feature vector, Y the ground truth, and U the sender's preferred outcome. Let $\ell : \mathbb{R} \times \mathbb{R} \to \mathbb{R}_+$ be the squared loss function given by $\ell(a, b) = (a - b)^2$.

The sender can choose to reveal the full feature vector X or nothing, denoted $\emptyset$. If the sender chooses to reveal the full feature vector X, it is transmitted through a noisy channel with a fixed, independent probability of erasure $q \in (0, 1)$. If the feature X survives and reaches the receiver, the receiver applies a linear regression model to X to make a prediction. If the receiver observes no feature, which may be due to the either sender choosing not to reveal it or due to erasure by the noisy channel, the receiver applies a default constant predictor. The sender's objective is to minimize the mean squared error (MSE) between the receiver's prediction and the sender's preferred outcome U, while the receiver's objective is to minimize the MSE between its prediction and the ground-truth Y.

We model this interaction as a Stackelberg game between the sender and the receiver, with the receiver as the leader. The receiver leads by committing to a two-tuple linear regression strategy $r_b \equiv (r_b(\emptyset), r_b(x))$ from the space of receiver strategies R, defined as[1]

$$\mathsf{R} := \{r_b : r_b(\emptyset) = 0,\ r_b(x) \equiv b^\top x, b \in \mathbb{R}^d\}.$$

This commitment occurs before the sender observes the tuple (U, X). The sender, knowing the receiver's committed strategy r_b, chooses a strategy A from the sender's strategy space, defined as

$$\mathsf{A} := \{A : \mathcal{Y} \times \mathcal{X} \to \mathcal{X} \cup \{\emptyset\} \mid A(u, x) \in \{x, \emptyset\}\ \forall (u, x)\}.$$

The sender's chosen action $A(U, X)$ is then communicated through a erasure channel parameterized by $q \in (0, 1)$ (erasure is independent of the random variables (U, X, Y)), defined as

$$M(A(u, x)) := \begin{cases} x & \text{with probability } 1 - q \text{ if } A(u, x) = x, \\ \emptyset & \text{otherwise.} \end{cases}$$

The output of the channel, denoted by M, is the message obtained by the receiver. The receiver then applies $r_b(M)$ to make a prediction. The sender incurs a loss of $\ell(u, r_b(M))$, while the receiver incurs a loss of $\ell(y, r_b(M))$.

The sender's objective is to minimize its MSE for a given receiver strategy r_b. It therefore chooses the best-response strategy $A^\star \in \mathsf{A}$ given by

$$A^\star(u, x) \equiv \arg \min_{A(u,x) \in \{x, \emptyset\}} \mathbb{E}_M[\ell(u, r_b(M(A(u, x))))].,$$

[1] The receiver's strategy space can be extended by allowing $r_b(\emptyset) = c_0$ and $r_b(X) = b^\top x + c$, where c_0, c are any constants. However, are result holds even we restrict the receiver to the smaller startegy space R considered above.

where we break the tie in favour of action $\emptyset$, and $\mathbb{E}_M[\cdot]$ denotes the expectation over the channel noise.

For a given receiver strategy r_b, under the sender's best-response $A^\star$, the receiver's MSE, referred to as the *strategic risk* and denoted by $R_S(b,q)$, is given by

$$R_S(b,q) := \mathbb{E}_{(U,X,Y,M)}[\ell(Y, r_b(M(A^\star(U,X;b,q))))].$$

The receiver's objective is to minimize its strategic risk $R_S(b,q)$. It therefore selects an optimal strategy $r_S^\star$ given by

$$r_S^\star := \arg\min_{r_b \in \mathsf{R}} R_S(b,q).$$

We denoted the minimum risk above by $R_S^*(q) = \widehat{R}_S(,q)$ and call it the *optimal strategic risk*. The joint sender–receiver strategy pair $(A^\star, r_S^\star)$ is called a *Stackelberg equilibrium* of the game. The full game structure is referred to as the environment, denoted by $\mathcal{E} := (U, X, Y, \mathsf{A}, \mathsf{R}, q)$.

We now pose two problems: the decision problem and the learning problem. In the decision problem the environment is known to the receiver and we derive conditions on the environment under which voluntary disclosure (also refered to as strategic disclosure) is superior to mandatory disclosure (also refered to as vanilla disclosure). In the learning problem, the random variable (U,X,Y) of the environment is unknown to the receiver, but finite i.i.d. samples from (U,X,Y) are available. We construct a learning algorithm that, with sufficient training, approaches the risk of the optimal model under the optimal disclosure regime (voluntary or mandatory) for that environment.

A. Decision problem

For a given known environment, we compare the risk of the voluntary disclosure regime with that of the mandatory disclosure regime. In the mandatory disclosure regime, the sender's action is fixed to always reveal; i.e., $A(u,x) \equiv x \in \mathsf{A}$. For any arbitrary strategy $r_b \in \mathsf{R}$, the receiver's MSE, referred to as the *vanilla risk* and denoted by $R_V(b,q)$ is

$$R_V(b,q) := (1-q)\mathbb{E}[\ell(Y, b^\top X)] + q\mathbb{E}[\ell(Y,0)].$$

The optimal vanilla strategy is given by

$$r_V^*(\emptyset) := \mathbb{E}[Y] = 0 \text{ and } r_V^*(x) := \mathbb{E}[Y|X=x] = b_V^{*\top} x,$$

where $b_V^* = \mathrm{Var}(X)^{-1}\mathrm{Cov}(X,Y)$ is the optimal slope for the linear regression model over (X,Y). Here $\mathrm{Cov}(X,Y) := \mathbb{E}[XY^\top]$ and $\mathrm{Var}(X) := \mathbb{E}[XX^\top]$. The *optimal vanilla risk* is denoted by $R_V^*(q)$. For the optimal slope $b = b_V^*$, the optimal vanilla risk is denoted by $R_V^*(q)$.

For a given environment, we define the strategic advantage, denoted by $\Delta(\mathcal{E})$, as

$$\Delta(\mathcal{E}) := R_V^*(q) - R_S^*(q).$$

The voluntary disclosure regime performs better if and only if the strategic advantage is positive. We derive a necessary condition and a sufficient condition on the structure of the environment for the strategic advantage to be positive.

B. Learning problem

In the learning problem, the distribution of the random variables (U,X,Y) is unknown to the receiver (we assume q is known), but the receiver has access to a finite number of i.i.d. samples from the environment. The objective here is to construct a learning algorithm for the receiver, that takes the dataset of samples as input and outputs both a strategy r_b and a disclosure regime (voluntary or mandatory) that the sender must adhere to. For example if $\mathcal{A}(\mathcal{S}) = (r_b, \text{voluntary})$, then the algorithm ask that the receiver make feature revelation voluntary for the sender, and commit to the strategy r_b. We compare the performance of the learning algorithm $\mathcal{A}$ with the risk of the optimal predictor under the optimal disclosure regime for that environment.

We assume throughout the learning problem setup that there exists a constant $C > 0$ such that the support of the random variable X, Y and U is contained in $[-C,C]^d$, $[-C,C]$ and $[-C,C]$ respectively. Furthermore, the receiver's strategy space consists of linear models with bounded slopes:

$$\mathsf{R} := \{r_b : r_b(\emptyset) = 0,\ r_b(x) = b^\top x,\ |b_i| \le C\}.$$

Let $\mathcal{S} = \{(u_i, x_i, y_i)\}_{i=1}^n$ be a dataset of n i.i.d. samples drawn from the distribution of (U,X,Y). For a given receiver strategy r_b, we define

$$\widehat{R}_S(b,q) := \frac{1}{n}\sum_{i=1}^n y_i^2 - (1-q)\frac{1}{n}\sum_{i=1}^n (2y_i - b^\top x_i) b^\top x_i \times \mathbb{I}_{\{(2u_i - b^\top x_i) b^\top x_i > 0\}},$$

$$\widehat{R}_V(b,q) := \frac{1}{n}\sum_{i=1}^n y_i^2 - (1-q)\frac{1}{n}\sum_{i=1}^n (y_i - b^\top x_i)^2,$$

where $\mathbb{I}\{\cdot\}$ is the indicator function. This unusual form of the empirical risk equals the empirical MSE, and it will become apparent in the proofs of Lemma 1 and Lemma 3. We refer $\widehat{R}_S(b,q)$ and $\widehat{R}_V(b,q)$ as the *empirical strategic risk* and *empirical vanilla risk* respectively. The corresponding empirical optimal predictors (parameterized by their slopes) are

$$\widehat{b}_V := \arg\min_b \widehat{R}_V(b,q), \quad \widehat{b}_S := \arg\min_b \widehat{R}_S(b,q).$$

Let $\mathcal{A} : \mathcal{S} \to \mathsf{R} \times \{\text{voluntary, mandatory}\}$ denote a learning algorithm that takes the dataset $\mathcal{S}$ as input and outputs both a strategy r_b and a disclosure regime. The excess risk of $\mathcal{A}$ on $\mathcal{S}$, denoted as $\mathcal{R}(\mathcal{A};\mathcal{S})$, is the gap between the risk of the strategy chosen by the algorithm under the regime chosen by the algorithm, and the minimum of the two optimal risks for each disclosure regime, i.e.,

$$\mathcal{R}(\mathcal{A};\mathcal{S}) := R_\theta(b;q) - \min(R_S^*(q), R_V^*(q)),$$

where θ is the regime chosen by $\mathcal{A}$. We construct $\mathcal{A}$ in Section III-B (Algorithm 1) and establish bounds on its excess risk in terms of the sample size n, the complexity of the strategy space R, and the bounding constant C.

III. Main Results

In this section, we establish results on both the decision and learning problems. For the decision problem, we establish a necessary condition and a sufficient condition on the environment under which the strategic advantage is positive. These results provide structural insights into the environmental parameters that yield a positive strategic advantage. We then perform simulations over a family of parameterized environments and visualize the strategic advantage as a function of the environmental parameters. We verify that parameters satisfying the sufficient condition lie within the region of positive strategic advantage. We then present our the learning problem and analyse its excess risk

A. Decision problem

In this section, we analyse the decision problem. We begin by establishing the vanilla risk.

Lemma 1. *Consider the mandatory disclosure setup. Let q be the channel noise parameter. The vanilla risk for a receiver strategy $r_b(\emptyset) = 0$, $r_b(x) = b^\top x$ is*

$$\begin{aligned} R_V(b,q) = \mathit{Var}(Y) - (1-q)\mathit{Var}({b_V^\star}^\top X) \\ + (1-q)\mathit{Var}((b-b_V^\star)^\top X), \end{aligned} \tag{1}$$

where $b_V^\star = \mathit{Var}(X)^{-1}\mathit{Cov}(X,Y)$ is the optimal slope. The optimal vanilla risk is

$$R_V^\star(q) = \mathit{Var}(Y) - (1-q)\mathit{Var}({b_V^\star}^\top X). \tag{2}$$

Proof. For any $r_b \in \mathsf{R}$, the vanilla risk expands as the expected loss over noisy and successful transmissions:

$$R_V(b,q) = q\mathbb{E}[Y^2] + (1-q)\mathbb{E}[(Y - b^\top X)^2]. \tag{3}$$

We decompose the second term by adding and subtracting ${b_V^\star}^\top x$ to get

$$\begin{aligned} \mathbb{E}[(Y - b^\top X)^2] = \mathbb{E}[(Y - {b_V^\star}^\top X)^2] + \mathbb{E}[({b_V^\star}^\top X - b^\top X)^2] \\ + 2\mathbb{E}[(Y - {b_V^\star}^\top X)({b_V^\star}^\top X - b^\top X)]. \end{aligned} \tag{4}$$

The cross term vanishes due to the orthogonality principle of linear least squares. Furthermore, $\mathbb{E}[({b_V^\star}^\top X)^2] = \mathrm{Var}({b_V^\star}^\top X)$ because $\mathbb{E}[{b_V^\star}^\top X] = 0$. Therefore, we get

$$\mathbb{E}[(Y - b^\top X)^2] = \mathrm{Var}(Y) - \mathrm{Var}({b_V^\star}^\top X) + \mathrm{Var}((b - b_V^\star)^\top X).$$

Combining the terms with their probabilities, we get (1) and setting $b = b_V^\star$ gives (2) as claimed. □

Consider the receiver strategy $r_b \in \mathsf{R}$. Associate with it the following functions:

$$F_b(u,x) := (2u - b^\top x)b^\top x, \quad G_b(x,y) := (2y - b^\top x)b^\top x. \tag{5}$$

We will use the shorthand G_b for $G_b(X,Y)$ and F_b for $F_b(U,X)$ when the random variables are clear from the context.

Lemma 2. *Consider a receiver strategy $r_b \in \mathsf{R}$. The sender's best response satisfies $A^\star(u,x) = x$ if and only if $F_b(u,x) > 0$.*

Proof. A sender with tuple (u,x) and given receiver strategy r_b prefers to disclose if and only if

$$\mathbb{E}_M[\ell(u, r_b(M(\emptyset)))] > \mathbb{E}_M[\ell(u, r_b(M(x)))].$$

Expanding the expectation over the channel noise, and rearranging the terms, and noting that $(1-q) < 0$, this simplifies to $F_b(u,x) = (2u - b^\top x)b^\top x > 0$. □

Remark 1. The function $F_b(u,x)$ determines the sender's strategic decision boundary. The function $G_b(x,y)$ measures the receiver's squared loss reduction, $y^2 - (y - b^\top x)^2$, when the feature is revealed and used in prediction.

We now calculate the strategic risk for a given receiver strategy $r_b \in \mathsf{R}$.

Lemma 3. *For a given receiver strategy $r_b \in \mathsf{R}$, the strategic risk is*

$$R_S(b,q) = \mathit{Var}(Y) - (1-q)\mathbb{E}\big[G_b(X,Y)\mathbb{I}\{F_b(U,X) > 0\}\big] \tag{6}$$

Proof. The strategic risk expands as the expected loss over the regimes of withholding and revealing:

$$\begin{aligned} R_S(b,q) &= \mathbb{E}\Big[\mathbb{I}\{F_b(u,x) \le 0\}Y^2 + \mathbb{I}\{F_b(u,x) > 0\}\big(qY^2 \\ &\quad + (1-q)(Y - b^\top X)^2\big)\Big] \\ &= \mathbb{E}[Y^2] - (1-q)\mathbb{E}\big[\mathbb{I}\{F_b > 0\}\big(Y^2 - (Y - b^\top X)^2\big)\big]. \end{aligned}$$

Thus,

$$R_S(b,q) = \mathrm{Var}(Y) - (1-q)\mathbb{E}\big[G_b(X,Y)\mathbb{I}\{F_b(U,X) > 0\}\big],$$

as claimed. □

We now evaluate a sufficient condition for the strategic advantage to be positive.

Theorem 1. *(Sufficient condition) For a given environment $\mathcal{E}$ with noise parameter q, the strategic advantage is positive if there exists $b \in \mathbb{R}^d$ such that*

$$\begin{aligned} \|U - Y\|_2 < \frac{\mathbb{E}\big[|G_b(X,Y)|\big] - \mathit{Var}({b_V^\star}^\top X)}{4\sqrt{\mathit{Var}(b^\top X)}} \\ - \frac{\mathit{Var}((b - b_V^\star)^\top X)}{4\sqrt{\mathit{Var}(b^\top X)}}, \end{aligned} \tag{7}$$

where $\|U - Y\|_2 = \sqrt{\mathit{Var}(U) + \mathit{Var}(Y) - 2\mathit{Cov}(U,Y)}$, $b_V^\star = \mathit{Var}(X)^{-1}\mathit{Cov}(X,Y)$, and $G_b(X,Y)$ given in (5).

Proof. Using Lemmas 1 and 3, the risk difference is:

$$\begin{aligned} R_S(b,q) - R_V^\star(q) = (1-q)\Big(\mathrm{Var}({b_V^\star}^\top X) - \mathbb{E}\big[G_b \\ \times \mathbb{I}\{F_b > 0\}\big]\Big). \end{aligned} \tag{8}$$

We rewrite the expectation as $\mathbb{E}[G_b\mathbb{I}\{F_b > 0\}] = \mathbb{E}[G_b] - \mathbb{E}[G_b\mathbb{I}\{F_b \le 0\}]$. Expanding $\mathbb{E}[G_b]$ gives $\mathbb{E}[G_b] = 2b^\top\mathrm{Var}(X)b_V^\star - \mathrm{Var}(b^\top X)$ Using the identiy $\mathrm{Var}((b - b_V^\star)^\top X) = \mathrm{Var}(b^\top X) - 2b^\top\mathrm{Var}(X)b_V^\star + \mathrm{Var}({b_V^\star}^\top X)$, we get $\mathbb{E}[G_b] = \mathrm{Var}({b_V^\star}^\top X) - \mathrm{Var}((b - b_V^\star)^\top X)$.

Next, we bound $\mathbb{E}[G_b\mathbb{I}\{F_b \leq 0\}]$. We decompose it as $\mathbb{E}[G_b\mathbb{I}\{F_b \leq 0\}] = \mathbb{E}[G_b\mathbb{I}\{G_b \leq 0\}] + \mathbb{E}[G_b(\mathbb{I}\{F_b \leq 0\} - \mathbb{I}\{G_b \leq 0\})] = -\frac{1}{2}\mathbb{E}[|G_b| - G_b] + \mathbb{E}[G_b\mathbb{I}\{\text{sign}(F_b) \neq \text{sign}(G_b)\}]$. When $\text{sign}(F_b) \neq \text{sign}(G_b)$, we have $|G_b| < |G_b - F_b|$. Since $G_b - F_b = 2(Y-U)b^\top X$, we get $|G_b - F_b| = 2|Y - U||b^\top X|$. Thus, applying Cauchy-Schwarz, we get $\mathbb{E}[G_b\mathbb{I}\{\text{sign}(F_b) \neq \text{sign}(G_b)\}] \leq 2\mathbb{E}[|Y - U||b^\top X|] \leq 2\|U - Y\|_2\sqrt{\text{Var}(b^\top X)}$. Substituting these components back into the risk difference equation gives

$$R_S(b,q) - R_V^\star(q) \leq (1-q)\Big(\frac{-\mathbb{E}[|G_b|] + \text{Var}({b_V^\star}^\top X)}{2} + \frac{\text{Var}((b - b_V^\star)^\top X)}{2} + 2\|U - Y\|_2\sqrt{\text{Var}(b^\top X)}\Big). \tag{9}$$

For $R_S(b,q) < R_V^\star(q)$, it suffices that the RHS is strictly negative, which rearranges directly to the stated inequality (7). □

Observe that the sufficient condition does not depend on the noise parameter q. The factor $(1-q)$ only scales the strategic advantage and never changes its sign.

We now evaluate a necessary condition for the strategic advantage to be positive.

Theorem 2. *(Necessary condition) For a given environment with noise parameter q, a necessary condition for the strategic advantage to be positive is that there exists $b \in \mathbb{R}^d$ such that*

$$\mathbb{E}\big[|G_b(X,Y)|\big] - \mathit{Var}({b_V^\star}^\top X) - \mathit{Var}((b - b_V^\star)^\top X) > 0, \tag{10}$$

where $b_V^\star = \mathit{Var}(X)^{-1}\mathit{Cov}(X,Y)$ and $G_b(X,Y)$ is defined as in (5).

Proof. Substituting $\mathbb{E}[G_b]$ into (8), we can rewrite the exact risk difference as

$$R_S(b,q) - R_V^\star(q) = (1-q)\Big(\text{Var}((b - b_V^\star)^\top X) + \mathbb{E}[G_b\mathbb{I}\{F_b \leq 0\}]\Big).$$

Note that $\mathbb{E}[G_b\mathbb{I}\{F_b \leq 0\}] \geq \mathbb{E}[G_b\mathbb{I}\{G_b \leq 0\}]$. Using $\mathbb{E}[G_b\mathbb{I}\{G_b \leq 0\}] = -\frac{1}{2}\mathbb{E}[|G_b| - G_b]$, we obtain:

$$\begin{aligned} R_S(b,q) - R_V^\star(q) &\geq (1-q)\Big(\text{Var}((b - b_V^\star)^\top X) + \frac{\mathbb{E}[G_b] - \mathbb{E}[|G_b|]}{2}\Big) = (1-q)\Big(\text{Var}((b - b_V^\star)^\top X) \\ &+ \frac{\text{Var}({b_V^\star}^\top X) - \text{Var}((b - b_V^\star)^\top X) - \mathbb{E}[|G_b|]}{2}\Big) \\ &= (1-q)\Big(\frac{-\mathbb{E}[|G_b|] + \text{Var}({b_V^\star}^\top X) + \text{Var}((b - b_V^\star)^\top X)}{2}\Big). \end{aligned}$$

The RHS is non-negative whenever $-\mathbb{E}[|G_b|] + \text{Var}({b_V^\star}^\top X) + \text{Var}((b - b_V^\star)^\top X) \geq 0$. Hence, the contrapositive provides the stated necessary condition for $R_S(b,q) - R_V^\star(q) < 0$. □

The necessary condition (10) requires the numerator on the right-hand side of the sufficient condition (7) to be strictly positive. This is intuitive, if it fails, even the case $U = Y$ yields no strategic advantage. Once the necessary condition holds, the left-hand side of the sufficient condition requires only that the alignment between the preferred valuation U and the ground truth Y is sufficiently tight.

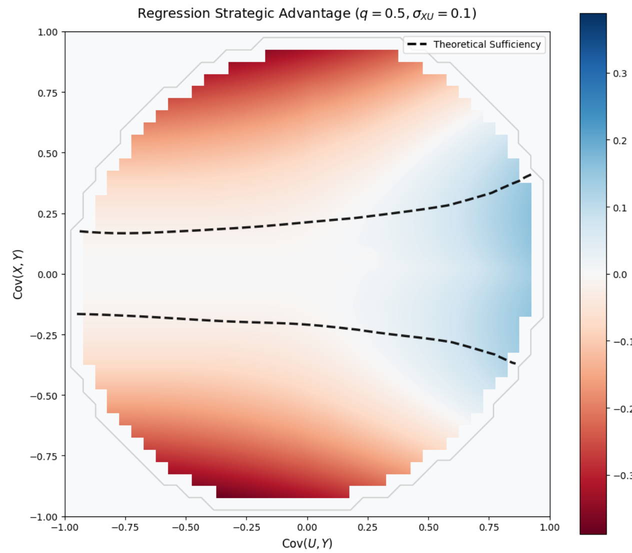

Fig. 1. Heatmap of the strategic advantage ($R_V^\star - R_S^\star$) for regression ($q = 0.5$, Cov(X,U)=0.1). Horizontal axis: Cov(U,Y), vertical axis: Cov(X,Y). The black dotted line is the theoretical sufficiency boundary (Theorem 1). Blue regions indicate parameter regimes where the voluntary disclosure regime is better.

To verify our results, we simulate a zero-mean multivariate Gaussian environment (U,X,Y) with unit variances, setting $q = 0.5$ and $\text{Cov}(X,U) = 0.1$. We estimate the strategic advantage on a 40×40 grid with $\text{Cov}(U,Y), \text{Cov}(X,Y) \in [-1,1]$, using $n = 10^5$ samples for the heatmap and for the sufficiency boundary. Figure 1 shows the strategic advantage as a function of the parameters $\text{Cov}(U,Y)$ and $\text{Cov}(X,Y)$, and confirms that the theoretical sufficiency region lies within the positive advantage (blue) zone, validating Theorem 1.

B. Learning problem

In the learning problem, the distribution of the random variable (U,X,Y) is unknown, but we have access to a finite number of samples from it. In this section, we present the following learning algorithm and establish bounds on its excess risk.

We now analyze the excess risk of the algorithm $\mathcal{A}$. We begin by introducing the necessary definitions and notation. The *empirical strategic advantage*, denoted as $\widehat{\Delta}(\mathcal{S})$, is defined as:

$$\widehat{\Delta}(\mathcal{S}) := \min_b \widehat{R}_V(b,q) - \min_b \widehat{R}_S(b,q).$$

Let $\mathcal{F}$ be a function class, and let $\mathcal{Z} := \{z_1, \ldots, z_n\}$ be a collection of points in the domain of $\mathcal{F}$. The *empirical Rademacher complexity* of $\mathcal{F}$ over $\mathcal{Z}$, denoted by $\widehat{\mathcal{R}}(\mathcal{F};\mathcal{Z})$, is defined as

$$\widehat{\mathcal{R}}(\mathcal{F};\mathcal{Z}) := \mathbb{E}_\sigma\left[\sup_{f \in \mathcal{F}} \frac{1}{n}\sum_{i=1}^n \sigma_i f(z_i)\right],$$

Algorithm 1 Joint learning algorithm for strategy and disclosure regime

Require: : Dataset $\mathcal{S} = \{(u_i, x_i, y_i)\}_{i=1}^n$ and strategy space $\widehat{\mathcal{R}} := \{\hat{r}_b : \hat{r}_\emptyset = 0,\ \hat{r}_x = b^\top x,\ |b_i| < C\}$ for some fixed constant $C > 0$.

▷ Solve empirical objectives

$\widehat{b}_S \leftarrow \arg\min_b \widehat{R}_S(b, q)$

$\widehat{b}_V \leftarrow \arg\min_b \widehat{R}_V(b, q)$

▷ Compute empirical strategic risk

$\widehat{R}_S(\widehat{b}_S, q) \leftarrow \frac{1}{n}\sum_{i=1}^n y_i^2 - (1-q)\frac{1}{n}\sum_{i=1}^n \big((2y_i - \widehat{b}_S^\top x_i)\widehat{b}_S^\top x_i\, \mathbb{I}\{(2u_i - \widehat{b}_S^\top x_i)\widehat{b}_S^\top x_i > 0\}\big)$

▷ Compute empirical vanilla risk

$\widehat{R}_V(\widehat{b}_V, q) \leftarrow \frac{1}{n}\sum_{i=1}^n y_i^2 - (1-q)\frac{1}{n}\sum_{i=1}^n (y_i - \widehat{b}_V^\top x_i)^2$

if $\widehat{R}_S(\widehat{b}_S, q) < \widehat{R}_V(\widehat{b}_V, q)$ **then**

  output $(\widehat{b}_S,$ voluntary$)$

else

  output $(\widehat{b}_V,$ mandatory$)$

end if

where σ_i are i.i.d. Rademacher random variables. We denote by $d_{\mathrm{VC}}(\mathcal{F})$ the VC dimension of the function class $\mathcal{F}$. For two function classes $\mathcal{F}$ and $\mathcal{G}$, the product class is defined as $\mathcal{F}\cdot\mathcal{G} := \{f\cdot g : f \in \mathcal{F}, g \in \mathcal{G}\}$, where $(f\cdot g)(z) = f(z)g(z)$ for all $z \in \mathbb{R}^d$. Furthermore, for a function $\phi : \mathbb{R} \to \mathbb{R}$, we define the composition class $\phi \circ \mathcal{G} := \{\phi \circ g : g \in \mathcal{G}\}$. If $\mathcal{F} = \phi \circ \mathcal{G}$, then for every $f \in \mathcal{F}$, there exists a $g \in \mathcal{G}$ such that $f(z) = \phi(g(z))$ for all $z \in \mathbb{R}^d$.

We now state some standard results in learning theory that we will use in our analysis.

Lemma 4. *Let $\mathcal{F} \subseteq \{f : \mathbb{R}^d \to \mathbb{R}\}$ and $\mathcal{G} \subseteq \{g : \mathbb{R}^d \to \mathbb{R}\}$ be function classes, and let $\mathcal{Z} = \{z_1, \ldots, z_n\}$ be a collection of n points in $\mathbb{R}^d$. Let $d_{VC}(\cdot)$ denote the VC dimension and let $\delta \in (0, 1)$.*

(i) For all $f \in \mathcal{F}$, with probability at least $1-\delta$, we have

$$\mathbb{E}[f(z)] \le \frac{1}{n}\sum_{\mathcal{Z}} f(z_i) + 2\widehat{\mathcal{R}}(\mathcal{F}; \mathcal{Z}) + 4C\sqrt{\frac{2\log(4/\delta)}{n}}.$$

(ii) If $\mathcal{F} = \{\phi \circ g : g \in \mathcal{G}\}$ where ϕ is L-Lipschitz, then $\widehat{\mathcal{R}}(\mathcal{F}; \mathcal{Z}) \le L\widehat{\mathcal{R}}(\mathcal{G}; \mathcal{Z})$.

(iii) If $\sup_{x, f\in\mathcal{F}} |f(x)| \le C_1$ and $\sup_{x, g\in\mathcal{G}} |g(x)| \le C_2$, then $\widehat{\mathcal{R}}(\mathcal{F}\cdot\mathcal{G}; \mathcal{Z}) \le C_2\widehat{\mathcal{R}}(\mathcal{F}; \mathcal{Z}) + C_1\widehat{\mathcal{R}}(\mathcal{G}; \mathcal{Z})$.

(iv) The Rademacher complexity and VC dimension satisfy

$$\widehat{\mathcal{R}}(\mathcal{F}; \mathcal{Z}) \le \mathcal{O}\Big(\sqrt{\frac{d_{VC}(\mathcal{F})\log n}{n}}\Big).$$

(v) Suppose the function class $\mathcal{F} := \{f : f(x) = \mathbb{I}\{Q(x) \ge 0\},\ Q(x) = x^\top A x + b^\top x + c,$ where $A \in \mathbb{R}^{d\times d}, b \in \mathbb{R}^d,$ and $c \in \mathbb{R}\}$. Then $d_{VC}(\mathcal{F}) \le \frac{(d+1)(d+2)}{2}$, where d is the dimension of the feature space.

(vi) Let $\mathcal{F} := \{f : f(x) = a + b^\top x, x \in [-C, C], a \in [-C, C], b \in [-C, C]^d\}$ be a linear function class. Then $\widehat{\mathcal{R}}(\mathcal{F}; \mathcal{Z}) = \frac{dC^2}{\sqrt{n}}$.

(vii) Let Z be a random variable with $\mathbb{E}[Z] = 0$ and $|Z| \le C$ a.s. Then with probability at least $1-\delta$,

$$\left|\frac{1}{n}\sum_{i=1}^n z_i^2 - \mathrm{Var}(Z)\right| \le C^2\sqrt{\frac{\log(2/\delta)}{n}}.$$

(viii) Let $\hat{f} \in \arg\min_{f\in\mathcal{F}} \frac{1}{n}\sum_{i=1}^n f(z_i)$ and $f^\star \in \arg\min_{f\in\mathcal{F}} \mathbb{E}[f(z)]$. Then with probability at least $1-\delta$

$$\mathbb{E}[\hat{f}(z)] \le \mathbb{E}[f^\star(z)] + 2\widehat{\mathcal{R}}(\mathcal{F}; \mathcal{Z}) + \mathcal{O}\left(C\sqrt{\frac{\log(1/\delta)}{n}}\right).$$

Proof. We refer the reader to the standard literature for the detailed proofs of the established claims. Claims (i) and (ii) follow directly from Theorem 26.5(ii) (p. 378) and Lemma 26.9 (p. 381) in Chapter 26 of [14], respectively. Claim (iii) follows by applying the contraction Lemma (Claim (ii)) as follows. For any $f \in \mathcal{F}$ bounded by C_1 and $g \in \mathcal{G}$ bounded by C_2, the product mapping $\phi(x, y) = x \cdot y$ is Lipschitz. By fixing one function and bounding the variation of the other, the Lipschitz constant with respect to f is bounded by $\sup|g| \le C_2$, and with respect to g is bounded by $\sup|f| \le C_1$. Applying the contraction property yields $\widehat{\mathcal{R}}(\mathcal{F}\cdot\mathcal{G}; \mathcal{Z}) \le C_2\widehat{\mathcal{R}}(\mathcal{F}; \mathcal{Z}) + C_1\widehat{\mathcal{R}}(\mathcal{G}; \mathcal{Z})$. Claim (iv) is derived by combining Sauer's Lemma (Corollary 3.3 (p. 47) Chapter 3 in [15]) and Massart's Lemma (Theorem 3.3 (p. 39) Chapter 3 in [15]). Sauer's Lemma bounds the growth function $\Pi_{\mathcal{F}}(n)$ for $n \ge d_{\mathrm{VC}}(\mathcal{F})$ as $\Pi_{\mathcal{F}}(n) \le (en/d_{\mathrm{VC}}(\mathcal{F}))^{d_{\mathrm{VC}}(\mathcal{F})}$. Massart's Lemma bounds the empirical Rademacher complexity in terms of this growth function as $\widehat{\mathcal{R}}(\mathcal{F}; \mathcal{Z}) \le C\sqrt{\frac{2\log\Pi_{\mathcal{F}}(n)}{n}}$. Substituting the bound for the growth function yields $\widehat{\mathcal{R}}(\mathcal{F}; \mathcal{Z}) \le \mathcal{O}\left(\sqrt{\frac{d_{\mathrm{VC}}(\mathcal{F})\log n}{n}}\right)$. Claim (v) is established using the polynomial embedding trick. A quadratic function $Q(x) = x^\top A x + b^\top x + c$ in $\mathbb{R}^d$ can be expressed as a linear inner product $\langle\tilde{w}, \tilde{x}\rangle$ in a lifted feature space. The lifted vector $\tilde{x}$ contains d linear terms x_i, $\frac{d(d+1)}{2}$ quadratic cross-terms $x_i x_j$ (for $i \le j$), and 1 constant term. The dimension of this new space is

$$D = \frac{d(d+1)}{2} + d + 1 = \frac{d^2 + 3d + 2}{2} = \frac{(d+1)(d+2)}{2}.$$

Since linear halfspaces in $\mathbb{R}^D$ have a VC dimension equal to D, the bound on $d_{\mathrm{VC}}(\mathcal{F})$ follows. Claim (vi) follows directly from Theorem 26.10 (p. 382) in [14]. Claim (vii) is a standard consequence of Hoeffding's concentration inequality. Since $|Z| \le C$ almost surely, the squared random variable Z^2 is bounded in the interval $[0, C^2]$. Applying Hoeffding's inequality to the empirical mean of Z^2 gives

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{i=1}^n z_i^2 - \mathbb{E}[Z^2]\right| > \epsilon\right) \le 2\exp\left(-\frac{2n\epsilon^2}{C^4}\right).$$

Setting the right-hand side equal to δ and solving for ϵ yields the stated bound. Finally, Claim (viii) follows directly from Theorem 26.5(iii) (p. 378) in [14]. □

We now establish a generalization bound for the vanilla risk.

Lemma 5. *Let $\mathcal{S}=\{(u_i,x_i,y_i)\}_{i=1}^n$ be a dataset of n i.i.d. samples. Let $r_b \in \mathsf{R}$ be any receiver strategy. Let $\delta \in (0,1)$. Then with probability at least $1-\delta$, we have*

$$R_V(b,q) - \widehat{R}_V(b,q) \leq \mathcal{O}\Big(\frac{d^2C^4}{\sqrt{n}} + C^2\sqrt{\frac{\log(1/\delta)}{n}}\Big).$$

Proof. Let $\hat{L} = \frac{1}{n}\sum_{\mathcal{S}}(y_i - r_b(x_i))^2$ and $L = \mathbb{E}[(Y - r_b(X))^2]$. We can rewrite the generalization gap as

$$R_V(b,q) - \widehat{R}_V(b,q) \leq |\mathrm{Var}(Y) - \frac{1}{n}\sum_{\mathcal{S}} y_i^2| + (1-q)(L-\hat{L}).$$

Applying a union bound, we allocate a failure probability of $\delta/2$ to each term. From Lemma 4 (vii), the variance estimation error is bounded by $\mathcal{O}\Big(C^2\sqrt{\log(1/\delta)/n}\Big)$.

We now bound $L - \hat{L}$. Let $\mathcal{H} := \{h_b(x,y) : h_b(x,y) = y - b^\top x, x \in \mathbb{R}^d, b \in \mathbb{R}^d, y \in \mathbb{R}\}$ be a linear function class. Recall that $\ell(a,b) = (a-b)^2$ is a quadratic loss function. Because $Y \in [-C, C]$ and $r_b(X) = b^\top X \in [-dC^2, dC^2]$, the maximum derivative of $\ell(y, b^\top x)$ is $2(dC^2 + C) = 2C(dC+1)$. Thus, ℓ is $2C(dC+1)$-Lipschitz. By 4 (ii) and Lemma 4 (vi), we have $\widehat{\mathcal{R}}(\ell \circ \mathcal{H}; \mathcal{Z}) \leq 2C(dC+1)\frac{dC^2}{\sqrt{n}} = \mathcal{O}\Big(\frac{d^2C^4}{\sqrt{n}}\Big)$.

Applying the Rademacher generalization bound from Lemma 4 (i), the term $L - \hat{L}$ is bounded by $2\widehat{\mathcal{R}}(\ell \circ \mathcal{H}; \mathcal{Z}) + \mathcal{O}\Big(C^2\sqrt{\log(1/\delta)/n}\Big)$. Combining the variance and loss terms yields the desired $\mathcal{O}\Big(\frac{d^2C^4}{\sqrt{n}} + C^2\sqrt{\frac{\log(1/\delta)}{n}}\Big)$ bound. □

Notice that in Lemma 5, the RHS does not depend on the choice of b and hence this bound is uniform over the strategy class R. Next, we establish a similar generalization bound for the strategic risk.

Lemma 6. *Let $\mathcal{S}=\{(u_i,x_i,y_i)\}_{i=1}^n$ be a dataset of n i.i.d. samples. Let $r_b \in \mathsf{R}$ be a receiver strategy. Let $\delta \in (0,1)$. Then with probability at least $1-\delta$, we have*

$$R_S(b,q) - \widehat{R}_S(b,q) \leq \mathcal{O}\Big(d^3C^4\sqrt{\frac{\log n}{n}} + C^2\sqrt{\frac{\log(1/\delta)}{n}}\Big). \tag{11}$$

Proof. Applying the triangle inequality, the difference $R_S(b,q) - \widehat{R}_S(b,q)$ can be upper bounded as

$$\begin{aligned} R_S(b,q) - \widehat{R}_S(b,q) &\leq |\mathrm{Var}(Y) - \frac{1}{n}\sum_{i\in\mathcal{S}} y_i^2| + (1-q) \\ &\times \Big|\mathbb{E}\big[G(X,Y)\mathbb{I}\{F(U,X)>0\}\big] \\ &- \frac{1}{n}\sum_{i\in\mathcal{S}} \mathbb{I}\{F(u_i,x_i)>0\}\, G(x_i,y_i)\Big|. \end{aligned}$$

Applying a union bound, we allocate a failure probability of $\delta/2$ to the variance term and $\delta/2$ to the expectation term. The variance term is bounded by $\mathcal{O}\Big(C^2\sqrt{\log(1/\delta)/n}\Big)$ from Lemma 4 (vii).

To bound the expectation term uniformly, we define the hypothesis classes $\mathcal{F} := \{f_b : f_b(u,x) = \mathbb{I}\{(2u - b^\top x)b^\top x > 0\}, b \in \mathbb{R}^d\}$ and $\mathcal{G} := \{g_b : g_b(x,y) = (2y - b^\top x)b^\top x, b \in \mathbb{R}^d\}$. Utilizing the Rademacher generalization bound from Lemma 4 (i), the gap is bounded by $2\widehat{\mathcal{R}}(\mathcal{F}\cdot\mathcal{G}; \mathcal{Z}) + \mathcal{O}\Big(C^2\sqrt{\log(1/\delta)/n}\Big)$.

To bound the Rademacher complexity of the product class $\mathcal{F}\cdot\mathcal{G}$, we break down the calculation and establish individual bounds for $\widehat{\mathcal{R}}(\mathcal{F}; \mathcal{Z})$ and $\widehat{\mathcal{R}}(\mathcal{G}; \mathcal{Z})$ in the following claims.

Claim 1. $\widehat{\mathcal{R}}(\mathcal{F}; \mathcal{Z}) \leq \mathcal{O}\Big(d\sqrt{\frac{\log n}{n}}\Big)$.

Proof of Claim. From Lemma 4 (iv), $\widehat{\mathcal{R}}(\mathcal{F}; \mathcal{Z}) \leq \mathcal{O}\Big(\sqrt{\frac{d_{\mathrm{VC}}(\mathcal{F})\log n}{n}}\Big)$. Because the decision boundary of $\mathcal{F}$ is quadratic in the feature space $\mathbb{R}^{d+1}$, from Lemma 4 (v), $d_{\mathrm{VC}} \leq \frac{(d+2)(d+3)}{2} = \mathcal{O}(d^2)$. Substituting this quadratic bound into the square root yields $\sqrt{\mathcal{O}(d^2)\log(n)/n}$, which simplifies to $\mathcal{O}\Big(d\sqrt{\log n/n}\Big)$. ■

Claim 2. $\widehat{\mathcal{R}}(\mathcal{G}; \mathcal{Z}) \leq \mathcal{O}\Big(\frac{d^2C^4}{\sqrt{n}}\Big)$.

Proof of Claim. Let $\phi_i(z) := (2y_i - z)z$ and $\mathcal{H} := \{h_b : h_b(x) = b^\top x, b \in \mathbb{R}^d\}$. From the definition of $\mathcal{G}$, $g_b(x_i,y_i) = \phi_i(h_b(x_i))$. Since $|Y| \leq C$ and $|Z| = |b^\top X| \leq dC^2$, the maximum derivative of $\phi_i(z)$ is bounded by $2C + 2dC^2 \leq 2C(1+dC)$. Thus, the function $\phi_i(z)$ is $2C(1+dC)$-Lipschitz. Applying Lemma 4 (ii) yields $\widehat{\mathcal{R}}(\mathcal{G}; \mathcal{Z}) \leq 2C(1+dC)\widehat{\mathcal{R}}(\mathcal{H}; \mathcal{Z})$. Since $\mathcal{H}$ is a linear class, $\widehat{\mathcal{R}}(\mathcal{H}; \mathcal{Z}) \leq \frac{dC^2}{\sqrt{n}}$, and the claim holds. ■

Because functions in $\mathcal{F}$ are bounded by 1, and functions in $\mathcal{G}$ are bounded by $\mathcal{O}(d^2C^4)$, from Lemma 4 (iii), we get $\widehat{\mathcal{R}}(\mathcal{F}\cdot\mathcal{G}; \mathcal{Z}) \leq \mathcal{O}(d^2C^4)\widehat{\mathcal{R}}(\mathcal{F}; \mathcal{Z}) + 1\cdot\widehat{\mathcal{R}}(\mathcal{G}; \mathcal{Z})$. Substituting the individual bounds yields $\mathcal{O}\Big(d^3C^4\sqrt{\frac{\log n}{n}}\Big)$. Combining the Rademacher complexity with the concentration terms completes the proof. □

From Lemma 5 and Lemma 6, we obtain the uniform generalization bound on the strategic advantage.

Theorem 3. *Let $\mathcal{S}=\{(u_i,x_i,y_i)\}_{i=1}^n$ be a dataset of n i.i.d. samples drawn from the random variable (U,X,Y) over $\mathcal{Y}\times\mathcal{X}\times\mathcal{Y}$. Let $\delta \in (0,1)$. Then with probability at least $1-\delta$, we have*

$$|\Delta(\mathcal{E}) - \widehat{\Delta}(\mathcal{S})| \leq \mathcal{O}\Big(d^3C^4\sqrt{\frac{\log n}{n}} + C^2\sqrt{\frac{\log(1/\delta)}{n}}\Big).$$

Proof. The absolute difference $|\Delta(\mathcal{E}) - \widehat{\Delta}(\mathcal{S})|$ is upper bounded by the triangle inequality

$$|R_V^\star(q) - \widehat{R}_V(\widehat{b}_V, q)| + |R_S^\star(q) - \widehat{R}_S(\widehat{b}_S, q)|.$$

We upper bound each of these terms using Lemma 4 (viii). Note that the right-hand side of this lemma is of the same order as Lemma 4 (i). Thus we apply the bounds derived in Lemma 5 for the vanilla risk term and Lemma 6 for the strategic risk term. We then take a union bound of $\delta/2$ for each term and absorb the lower-order vanilla risk term into the strategic risk term to arrive at the final bound.

□

We now establish the excess risk bound on the joint selection learning algorithm.

Theorem 4. *Consider Algorithm 1 and dataset $\mathcal{S}$. Let $\delta \in (0,1)$. Then with probability at least $1-\delta$, the excess risk of the algorithm is bounded by*

$$\mathcal{R}(\mathcal{A},\mathcal{S}) \le \mathcal{O}\Big(d^3C^4\sqrt{\frac{\log n}{n}} + C^2\sqrt{\frac{\log(1/\delta)}{n}}\Big).$$

Proof. Let $\epsilon(n,\delta) = \mathcal{O}\Big(d^3C^4\sqrt{\frac{\log n}{n}} + C^2\sqrt{\frac{\log(1/\delta)}{n}}\Big)$. By a union bound over the preceding lemmas, the generalization errors for both disclosure regimes and the advantage estimation error $|\Delta(\mathcal{E}) - \widehat{\Delta}(\mathcal{S})|$ are simultaneously bounded by $\epsilon(n,\delta)$ with probability at least $1-\delta$.

If the empirical advantage satisfies $|\widehat{\Delta}(\mathcal{S})| > \epsilon(n,\delta)$, the estimation error is insufficient to flip the sign of the advantage. Consequently, the algorithm identifies the optimal disclosure regime with certainty. In this scenario, the total excess risk $\mathcal{R}(\mathcal{A},\mathcal{S})$ is bounded solely by the generalization gap of the chosen regime, which is at most $\epsilon(n,\delta)$.

Otherwise, if $|\widehat{\Delta}(\mathcal{S})| \le \epsilon(n,\delta)$, the algorithm may select the suboptimal disclosure regime. This selection incurs an additional penalty equal to the true strategic advantage $|\Delta(\mathcal{E})|$. Since $|\Delta(\mathcal{E})| \le |\widehat{\Delta}(\mathcal{S})| + \epsilon(n,\delta) \le 2\epsilon(n,\delta)$, the total excess risk is bounded by the sum of the generalization error and the selection penalty, totaling $3\epsilon(n,\delta)$.

In both scenarios, the excess risk is $\mathcal{O}(\epsilon(n,\delta))$. Absorbing the constants yields the final bound as claimed. □

IV. Conclusion

In this work, we have established a necessary condition and a sufficient condition for a positive strategic advantage in linear regression. We further showed that the parameter of the noisy channel only scales the magnitude of the strategic advantage, but not its sign. The sufficient condition for a positive strategic advantage holds when the alignment between the ground-truth outcome and the preferred outcome is sufficiently high. We then proposed a joint learning algorithm that simultaneously learns the optimal disclosure regime and the optimal predictor for that regime. We showed that its excess risk is of order $\mathcal{O}\left(d^3C^4\sqrt{\frac{\log n}{n}} + C^2\sqrt{\frac{\log(1/\delta)}{n}}\right)$, which is the same order that any other algorithm would achieve if it knew the optimal disclosure regime a priori.

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